

EXECUTIVE SUMMARY

This report analyzes GPU benchmark results comparing 4 NVIDIA GPU power governors:

- **gpu_default** - Default NVIDIA driver behavior
- **gpu_ondemand** - Dynamically adjusts clock based on workload
- **gpu_performance** - Always runs at maximum clock
- **gpu_powersave** - Always runs at minimum clock (power saving)

KEY FINDINGS

1. POWER CONSUMPTION (Average across all benchmarks):

- **gpu_powersave**: 6.3W (75% less than default)
- **gpu_ondemand**: 27.5W (10% less than default)
- **gpu_default**: 30.6W (baseline)
- **gpu_performance**: 34.9W (14% more than default)

2. EXECUTION TIME (Average):

- **gpu_performance**: 5.9s (fastest)
- **gpu_default**: 5.9s
- **gpu_ondemand**: 6.6s (11% slower)
- **gpu_powersave**: 22.0s (272% slower!)

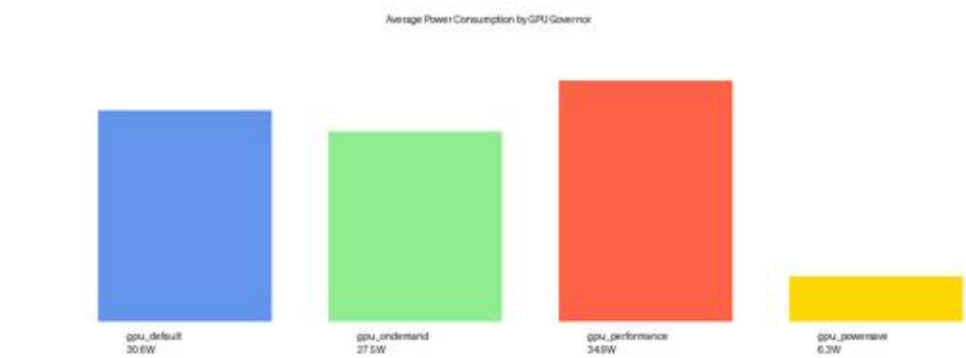
3. ENERGY EFFICIENCY (Power x Time = Joules):

- **gpu_powersave**: 150.7 J (BEST - 22% less energy)
- **gpu_default**: 192.5 J (baseline)
- **gpu_ondemand**: 203.3 J (6% more energy)
- **gpu_performance**: 214.1 J (11% more energy)

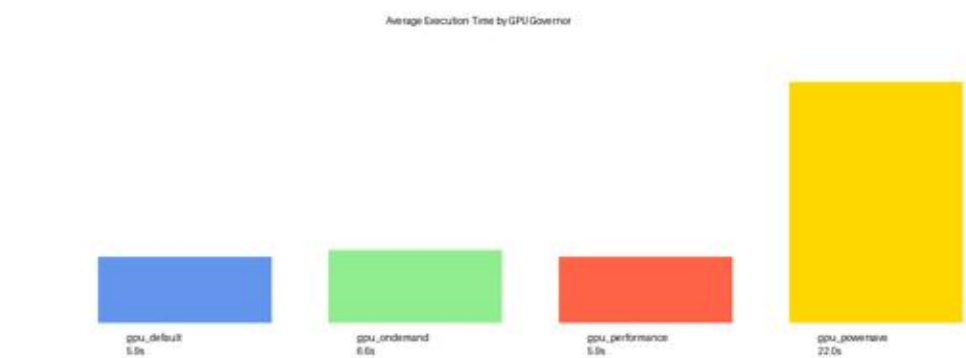
RECOMMENDATIONS

- **For MAXIMUM ENERGY SAVINGS:** Use **gpu_powersave**
(72% less power, but 3.7x slower)
- **For BALANCED** (good speed + power savings): Use **gpu_ondemand**
(10% less power, similar speed to default)
- **For MAXIMUM PERFORMANCE:** Use **gpu_performance**
(14% more power, but fastest execution)
- **For DEFAULT BEHAVIOR:** Use **gpu_default**
(good balance of speed and power)

Power Consumption by Governor



Execution Time by Governor



Energy Efficiency by Governor

